

Integrating using inverse hyperbolic trig

Basic skills

$$Q1. a) \operatorname{arctanh} x = \frac{1}{2} \ln \left(\frac{1+x}{1-x} \right)$$

$$b) \operatorname{arsinh} x = \ln (x + \sqrt{x^2 + 1})$$

$$c) \operatorname{arcosh} x = \ln (x \pm \sqrt{x^2 - 1})$$

$$Q2. \cosh x = s \text{ so } x = \cosh^{-1} s$$

$$\Rightarrow x = \ln (s \pm \sqrt{s^2 - 1}) = \ln (s \pm \sqrt{24})$$

$$Q3. y = \sinh^{-1} x$$

$$a) \frac{dy}{dx} = \frac{1}{\sqrt{x^2 + 1}}$$

$$b) x = \sinh y \Rightarrow \frac{dx}{dy} = \cosh y = \sqrt{1 + \sinh^2 y}$$

$$\Rightarrow \frac{dx}{dy} = \sqrt{1 + x^2} \text{ so } \frac{dy}{dx} = \frac{1}{\sqrt{x^2 + 1}}$$



Q4. a)
$$\int \frac{1}{\sqrt{x^2+25}} dx = \int \frac{1}{\sqrt{x^2+5^2}} dx$$

$$= \operatorname{arcsinh}\left(\frac{x}{5}\right) = \ln\left(\frac{x}{5} + \sqrt{\frac{x^2}{25} + 1}\right)$$

b)
$$\int \frac{1}{\sqrt{4x^2-200}} dx = \int \frac{1}{2\sqrt{x^2-50}} dx$$

$$= \frac{1}{2} \operatorname{arccosh}\left(\frac{x}{\sqrt{50}}\right) = \frac{1}{2} \ln\left(\frac{x}{\sqrt{50}} + \sqrt{\frac{x^2}{50} - 1}\right)$$

c)
$$\int \frac{1}{x^2-30} dx = \int \frac{-1}{30-x^2} dx$$

$$= -\operatorname{arctan}\left(\frac{x}{\sqrt{30}}\right) = -\frac{1}{2} \ln\left(\frac{1 + \frac{x}{\sqrt{30}}}{1 - \frac{x}{\sqrt{30}}}\right)$$

Q5. a)
$$\int_3^5 \frac{1}{\sqrt{3x^2-9}} dx = \int_3^5 \frac{1}{\sqrt{3}} \frac{1}{\sqrt{x^2-3}} dx$$

$$= \frac{1}{\sqrt{3}} \left[\cosh^{-1}\left(\frac{x}{\sqrt{3}}\right) \right]_3^5 = \frac{1}{\sqrt{3}} \left[\ln\left(\frac{x}{\sqrt{3}} + \sqrt{\frac{x^2}{3} - 1}\right) \right]_3^5$$

$$= \frac{1}{\sqrt{3}} \left(\ln\left(\frac{5}{\sqrt{3}} + \sqrt{\frac{24}{3}}\right) - \ln\left(\frac{3}{\sqrt{3}} + \sqrt{\frac{8}{3}}\right) \right) = \frac{1}{\sqrt{3}} \ln\left(\frac{5 + \sqrt{24}}{3 + \sqrt{8}}\right)$$

Law of logs and $\frac{x \sqrt{3}}{x \sqrt{3}}$

$$b) \int_0^1 \frac{s}{\sqrt{25x^2+1}} dx = \int_0^1 \frac{1}{\sqrt{x^2+(\frac{1}{5})^2}} dx$$

$$= \left[\sinh^{-1}\left(\frac{x}{\frac{1}{5}}\right) \right]_0^1 = \left[\sinh^{-1}(5x) \right]_0^1$$

$$= \left[\ln(5x + \sqrt{25x^2+1}) \right]_0^1 = \ln(5 + \sqrt{26}) - \ln(1)$$

$$= \ln(5 + \sqrt{26}) - 0 = \ln(5 + \sqrt{26})$$

$$c) \int_0^2 \frac{2}{20-2x^2} dx = \int_0^2 \frac{1}{10-x^2} dx$$

$$= \left[\tanh^{-1}\left(\frac{x}{\sqrt{10}}\right) \right]_0^2 = \left[\frac{1}{2} \ln\left(\frac{1+\frac{x}{\sqrt{10}}}{1-\frac{x}{\sqrt{10}}}\right) \right]_0^2$$

$$= \left[\frac{1}{2} \ln\left(\frac{\sqrt{10}+x}{\sqrt{10}-x}\right) \right]_0^2 = \frac{1}{2} \ln\left(\frac{\sqrt{10}+2}{\sqrt{10}-2}\right) - \frac{1}{2} \ln(1)$$

$$= \frac{1}{2} \ln\left(\frac{\sqrt{10}+2}{\sqrt{10}-2}\right)$$

$$Q6. \int \frac{1}{\sqrt{(3x-4)^2-25}} dx \quad \begin{array}{l} \text{Sub } u = 3x-4 \\ du = 3dx \end{array}$$

$$\text{so } \int \frac{1}{3} \frac{1}{\sqrt{u^2-5^2}} du = \frac{1}{3} \cosh^{-1}\left(\frac{u}{5}\right) \\ = \frac{1}{3} \cosh^{-1}\left(\frac{3x-4}{5}\right)$$

Q7.

$$\int \frac{1}{a^2 - x^2} dx = \int \frac{1}{a^2} \times \frac{1}{1 - \left(\frac{x}{a}\right)^2} dx$$

now let $u = \frac{x}{a} \Rightarrow du = \frac{1}{a} dx$

so $\int \frac{1}{a^2} \times \frac{1}{1 - u^2} \times a du = \frac{1}{a} \int \frac{1}{1 - u^2} du$
 $= a \operatorname{arctanh}^{-1}(u) = a \operatorname{arctanh}^{-1}\left(\frac{x}{a}\right)$

Advanced skills

complete square

Q8.

a) $\int_1^2 \frac{2}{\sqrt{x^2 - 4x + 5}} dx = \int_1^2 \frac{2}{\sqrt{(x-2)^2 + 1}} dx$

use $u = x - 2 \quad du = dx$

$= \int_{-1}^0 \frac{2}{\sqrt{u^2 + 1}} du = 2 \left[\sinh^{-1} u \right]_{-1}^0$

$= 2 \left[\ln(x + \sqrt{x^2 + 1}) \right]_{-1}^1 = 2 \left(\ln(1 + \sqrt{2}) - \ln(-1 + \sqrt{2}) \right)$

$= 2 \ln \left(\frac{1 + \sqrt{2}}{\sqrt{2} - 1} \right)$

$$b) \int \frac{1}{\sqrt{2x^2+4x-16}} dx = \int \frac{1}{\sqrt{2}\sqrt{(x+1)^2-9}} dx$$

$$\text{let } u = x+1 \quad du = dx$$

$$= \int \frac{1}{\sqrt{2}} \times \frac{1}{\sqrt{u^2-3^2}} du = \frac{1}{\sqrt{2}} \cosh^{-1}\left(\frac{u}{3}\right)$$

$$= \frac{1}{\sqrt{2}} \ln\left(\frac{u}{3} + \sqrt{\frac{u^2}{9} - 1}\right)$$

$$= \frac{1}{\sqrt{2}} \ln\left(\frac{x+1}{3} + \sqrt{\frac{(x+1)^2}{9} - 1}\right)$$

$$c) \int \frac{1}{20-6x-2x^2} dx = \frac{1}{2} \int \frac{1}{10-3x-x^2} dx$$

$$= \frac{1}{2} \int \frac{1}{12.25 - (x+1.5)^2} dx \quad \begin{array}{l} \text{Sub } u = x+1.5 \\ du = dx \end{array}$$

$$= \frac{1}{2} \int \frac{1}{12.25 - u^2} du = \frac{1}{2} \frac{1}{\sqrt{12.25}} \tanh^{-1}\left(\frac{u}{\sqrt{12.25}}\right)$$

$$= \frac{1}{2} \times \frac{1}{\sqrt{12.25}} \tanh^{-1}\left(\frac{x+1.5}{\sqrt{12.25}}\right)$$

$$= \frac{1}{4\sqrt{12.25}} \ln\left(\frac{1 + \frac{x+1.5}{\sqrt{12.25}}}{1 - \frac{x+1.5}{\sqrt{12.25}}}\right)$$

Q9. $\int \frac{1}{\sqrt{x^2+1}} dx$ let $x = \sinh u$
 $dx = \cosh u du$

$= \int \frac{1}{\sqrt{\sinh^2 u + 1}} \cosh u du$
 as $\cosh^2 u - \sinh^2 u = 1$

$= \int \frac{\cosh u}{\cosh u} du = \int 1 du = u = \sinh^{-1} x$

Q10. $\int \frac{1}{1-x^2} dx$ let $x = \tanh u$
 $dx = \operatorname{sech}^2 u du$

$= \int \frac{1}{1 - \tanh^2 u} \operatorname{sech}^2 u du$
 as $\tanh^2 u + \operatorname{sech}^2 u = 1$

$= \int \frac{\operatorname{sech}^2 u}{\operatorname{sech}^2 u} du = \int 1 du = u = \tanh^{-1} x$

Q11. $\int \sqrt{x^2-1} dx$ let $x = \cosh u$
 so $\cosh^2 x - 1 = \sinh^2 x$
 $\Rightarrow \sqrt{x^2-1} = \sqrt{\sinh^2 x} = \sinh x$
 $dx = \sinh u du$
 $u = \cosh^{-1} x$

now need to get multiple argument

$$\begin{aligned}\cosh 2x &= \cosh^2 x + \sinh^2 x \\ &= 1 + 2\sinh^2 x\end{aligned}$$

$$\text{so } \sinh^2 x = \frac{1}{2} + \frac{1}{2} \cosh 2x$$

$$\begin{aligned}\text{so } \int \sinh^2 u \, du &= \int \frac{1}{2} + \frac{1}{2} \cosh 2u \, du \\ &= \frac{1}{2} \left(u + \frac{1}{2} \sinh 2u \right) = \frac{1}{2} \left(u + \cosh u \sinh u \right)\end{aligned}$$

$\sinh 2x = 2 \cosh x \sinh x$

$$\cosh u = x$$

$$u = \cosh^{-1} x$$

$$\sinh u = \sqrt{x^2 - 1}$$

$$\begin{aligned}\text{so } &= \frac{1}{2} \left(\cosh^{-1} x + x \sqrt{x^2 - 1} \right) \\ &= \frac{1}{2} \left(\ln(x + \sqrt{x^2 - 1}) + x \sqrt{x^2 - 1} \right)\end{aligned}$$